

CE 310

The Flaw of Averages

Week 1 · Session 1 · Probability & Statistics in Civil and Architectural Engineering

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Why This Course Matters

Personal story

In the Age of AI

Importance of Probability and Statistics

I studied probability and statistics from high school onward — and for years I did not see why they mattered.

What I eventually understood is that this subject is a **lens**: it is how engineers look at an uncertain world and still make defensible decisions. That is why it is a required course.

You may build a very different career from mine. It will still come down to the same thing — the quality of the decisions you make with incomplete information.

Are you afraid that Artificial Intelligence (AI) will take over your jobs?

A new disruptive technology always arrives. The internet did. Large language models are doing it now. What does not change is the fundamentals — and there are only two places to stand.

WORKING WITH IT

You know what the tool is computing, so you can check it, catch it when it is wrong, and put your name on the result.

REPLACED BY IT

You can operate the tool, but you cannot tell a good answer from a bad one — so anyone who can, replaces you.

Which side you land on is decided by what you understand. That is the real reason this course is required: it teaches the part that does not go out of date.

AI Tools Are Already Part of Engineering

As an engineer, you might have used:

- Tools that **predict** how much energy a building will use next month
- Tools that **detect** when a sensor or structural component is behaving abnormally
- Tools that **estimate** cost ranges and construction schedules under uncertainty
- Tools that **recommend** design alternatives based on performance data

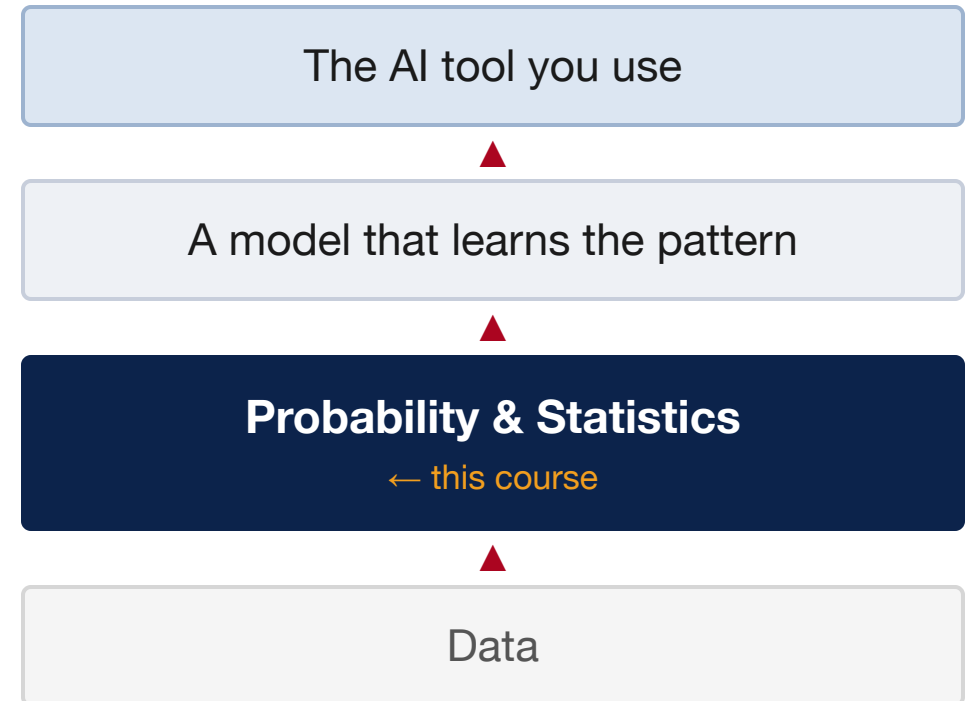
All of these tools take data and produce a prediction or a decision.

The question is: how do they actually work?

Probability and Statistics: The Foundation of All AI

Every AI tool learns from data. The math it uses to describe patterns, measure uncertainty, and make predictions is **probability and statistics** — nothing else.

What the AI tool does	The math underneath
Predicts next month's building energy use	Fits a curve to past data
Flags a sensor reading as suspicious	Checks against the normal range
Estimates how uncertain a cost forecast is	Simulates thousands of outcomes
Predicts the probability of a storm	Computes likelihood from records



The tools get fancier. The foundation stays the same.

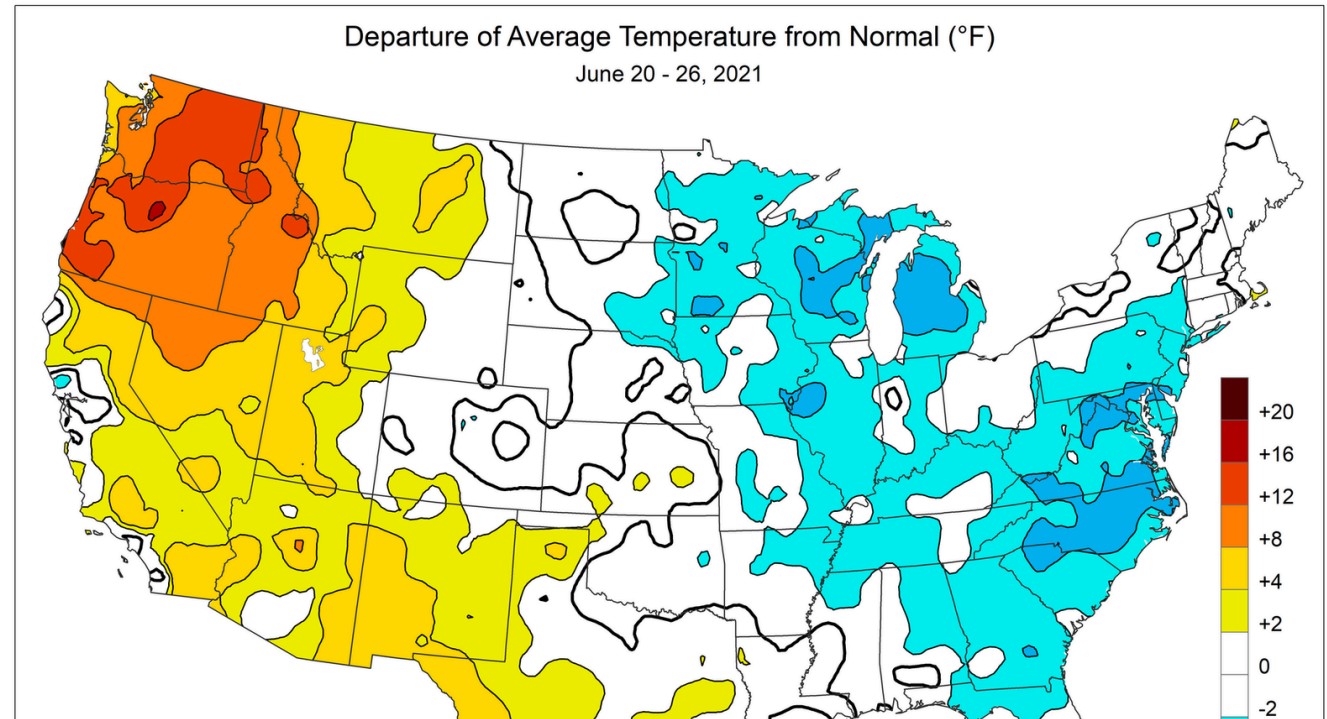
Why You Need to Understand It — Not Just Use It

You can use a calculator without understanding arithmetic — but not check its answer.

AI tools **fail silently** when conditions leave the range they were trained on.

June 2021: the Pacific Northwest ran **16–20 °F above normal**. Energy models built on historical climate files returned confident, wrong numbers.

Who catches that?



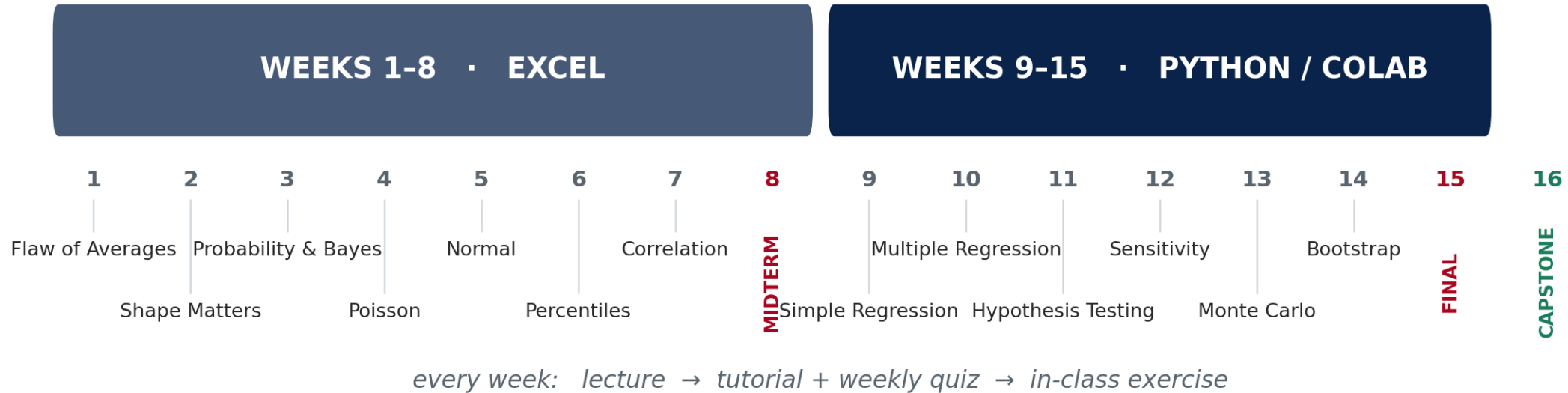
The 2021 heat dome, as a departure from the 1981–2010 normal — a statistic, not a temperature. Nothing in a model trained on that baseline had ever seen this. *NOAA Climate Prediction Center — public domain.*

You are actually in a good position.

AI has moved fastest in the digital world — text, images, code. **Physical AI** — systems that act on real buildings, roads and structures — is moving far more slowly.

Civil and architectural engineering is a field of physical assets. That gives you a longer runway than most professions. It will not last forever — Tesla is a preview of what integration looks like — but it is real, and it is yours to use.

How This Course Is Structured



Monday — theory lecture

Wednesday — tutorial · in-class exercise on a real dataset · Weekly Quiz opens

Friday — in-class exercise & assignments; finish at home

Assessments

Weekly Quizzes **5%** · In-Class Exercises **10%**

Homework **10%** · Midterm **25%**

Final **30%** · Capstone **15%**

Attendance **5%**

How to Use These Slides

These decks are written to be **re-read**, not read aloud. In class I spend the time on what is hard, or on what people get wrong; the rest is on the slide so you can rebuild it later without me. That is why they are denser than a talk — they are your study document, not my notes.

What the colours mean

 **Green** — a definition. The wording matters.

 **Blue** — the point of the slide.

 **Amber** — a mistake that costs marks or hours.

Two kinds of slide

Cream slides are your turn: a question with no answer on the page. The slide straight after works it through — so stop and try it before you scroll on.

Every deck ends with *Key Formulas* and *Key Results*. That is the revision summary. Start there when you study, then come back here for the reasoning.

Why Excel? — Weeks 1–8

It is universally used in engineering practice

Structural calculations, energy models, cost estimates, construction schedules — Excel is on every engineer's desktop, in every firm, at every agency. Knowing it well is a baseline expectation.

It is a gateway to programming thinking

Excel formulas already are programming:

- A function takes inputs and returns an output
- **IF** is conditional logic
- Filling a formula down a column is a loop

You are learning to think like a programmer — without the syntax burden. When you later write code in Python, the logic will feel familiar.

Why Python? — Weeks 9–15

It is the language of the AI era

It has become the shared language of data scientists, researchers, and engineers worldwide. I want you to be able to use it before you graduate. It is no longer optional.

It is designed to read like plain English

A single script can load a dataset, run a full statistical analysis, produce publication-quality charts, and build a prediction model — all in one place. Free, open source, and runs on any computer.

It has countless libraries

Statistics, optimization, mapping, image analysis, machine learning — there is a mature, free library for almost all of it. Excel has nothing comparable.

Excel → Python: The Path This Course Takes

	Excel	Python
When	Weeks 1–8	Weeks 9–15
You see	Every cell, every formula	Every line of logic
Best for	Building intuition, checking by hand	Large datasets, automation, AI
In practice	Daily tool in every engineering firm	The language of data science and AI

Excel teaches you *what* to compute and *why*. Python teaches you to do it at scale — the way real AI systems do.

You Already Know How to Program — MATLAB → Python

You've already taken the required MATLAB course. The core ideas transfer directly — you are not starting from zero in Week 9.

	MATLAB (prerequisite)	Python (Weeks 9–15)
Core ideas	Loops, conditionals, functions, vectors/matrices	Same ideas — different syntax
You already know	How to break a problem into logical steps	How to translate that logic into a new language
What's new	—	Reading real data files, statistics libraries, plotting library

Python isn't a new way of thinking for you — it's a new dialect for logic you already have.

Python tutorial materials are on D2L from day one — no need to wait.

Midterm and Final — What They Look Like

Both exams have the same two-part shape: one session on paper, one at the keyboard.

Midterm — Week 8 · 25%

Wed Oct 14 — Concept Exam

50 min · 50 pts · on paper

Pencil and calculator, nothing else

Fri Oct 16 — In-Class Exercise Practicum

50 min · 100 pts · open Excel, open notes

New dataset · no internet, no AI

Covers Weeks 1–7 — the Excel half

Final — Week 15 · 30%

Mon Nov 30 — Written Methods

50 min · 50 pts · on paper

Pencil and calculator, nothing else

Wed Dec 2 — Python Practicum

50 min · 100 pts · open notes, Colab only

New dataset · no internet, no AI

Covers Weeks 9–14 — the Python half

💡 The paper session asks whether you can reason about a method without a computer. The practicum asks whether you can make the tool produce a number you can defend.

The Capstone Project — Weeks 15–16 · 15%

 **Details to be determined.** The confirmed brief — team assignment, dataset rules, deliverables and dates — comes later in the semester.

What You Need — and How to Get It

Excel · Weeks 1–8

Free with your NetID. Microsoft 365 includes Excel and installs on your own machine — see it.arizona.edu.

Or use any campus computer lab: Excel is on every lab machine, nothing to install.

Python · Weeks 9–15

Google Colab runs in a browser. Nothing to install, works on a tablet.

A **local install** — Anaconda or VS Code — also works, and the Week 15 practicum needs one, because it runs with the Wi-Fi off.

No laptop? The Library lends them

Rhonda G. Tubbs Tech Toolshed — Main Library, 1st floor, or the Health Sciences Library desk. Bring your **CatCard**, in person, during open hours.

Free. The loan auto-renews every 21 days up to **147 days**, which covers this semester. One machine at a time.

They also lend **Wi-Fi hotspots**, calculators and more — 520-621-6442.

 A returned laptop is **wiped** — everything you installed and saved goes with it. Keep your work in OneDrive or submitted to D2L, never only on the machine.

Your Instructor

Wooyoung Jung, Ph.D.

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Education

Degree	Field	Institution	Year
Ph.D.	Civil Engineering	Virginia Tech	2020
M.S.	Architectural Engineering	Yonsei University	2016
B.S.	Architectural Engineering	Yonsei University	2011



What My Research Is About

Human-AI-Building Collaboration

— how humans, AI and buildings can work together to make buildings smarter, healthier and more efficient.

- How can people interact with buildings effectively?
- How can AI make those interactions easier?
- Physical AI in buildings

💡 Every project starts with exactly what this course teaches.



A building automation panel — the controller and field modules behind the sensor streams my lab works with. Temperature, occupancy and energy readings all start here, as raw numbers in a file, before anyone can say anything about them.

Interested in Building + AI?

Consider Joining my Human Building Synergy (HUBS) Lab

How it usually works: sit in with the lab for a semester, then register for research credit in later semesters (up to three credits) if it is a good fit for both sides.

What we work on: Building AI — how AI can work well with buildings *and* with the people in them.

Who I am looking for: students who are genuinely curious about this area and will follow through on what they commit to. Prior research experience is not required.

Check out my lab's webpage: <https://hubs.engr.arizona.edu/>

Where to Get Help

D2L Discussions — ask here first

Every week has its own topic covering both the in-class exercise and the homework. Post there rather than emailing me: one answer reaches the whole class, and somebody has usually asked already.

⚠ Methods, error messages, "why does this formula return zero" — yes. Your computed values or filled-in cells — no. That is sharing answers.

Prosper Babon-Ayeng · Teaching Assistant

Leads the Wednesday tutorial

Office hours: Mondays 10:00–11:00 AM · Civil Building 118

pbabonayeng@arizona.edu

Dr. Wooyoung Jung · Instructor

Office hours: by appointment — email to schedule

wooyoung@arizona.edu · Civil Building 220D

Replies within 48 hours on business days

Attendance — Set Up Top Hat Before Wednesday

Attendance is **5% of your grade** and is recorded automatically. The first check-in runs **Wednesday, Aug 26**.

Set it up once — in a web browser, not the mobile app

1. Go to **app.tophat.com**
2. Search *University of Arizona* → **Log in with school account**
3. Sign in with your **NetID and password** — not your email address
4. Choose **Sign up as a Student**. Already have a Top Hat account? Choose *Merge existing account with SSO*
5. Join this course with code **222566**

The account is **free**. Sign-up trouble goes to intech@arizona.edu or support@tophat.com — not to me, as I cannot fix accounts.

How the check-in works

There is a check-in at **every class from Wednesday, Aug 26**. The code appears on screen at a point I pick during the session — sometimes at the start, sometimes not, and never announced in advance. Enter it in Top Hat before it closes.

Location is on. Top Hat confirms your device is in this room, so a check-in from anywhere else does not register.

 Check in only for yourself. Checking in for an absent classmate is an integrity violation for both of you, handled under the UA Code rather than by an attendance deduction.

Present but the check-in did not register? Email me **within 24 hours** of that session.

Part 1

The Flaw of Averages

A Famous Engineering Failure

In the 1950s, the U.S. Air Force measured **10 physical dimensions** on **4,063 airmen** — height, arm reach, hip width, and seven others.

They designed cockpits to fit the pilot with average measurements on all 10 dimensions.



F-86 Sabre cockpit, the fighter-jet era of the 1950 Air Force study. U.S. Air Force / National Museum of the USAF — public domain.

How many of the 4,063 pilots fit the "average" cockpit?

Zero.

Not one pilot was average on all 10 dimensions simultaneously.

Designing for the average meant the cockpit fit nobody.

The fix — adjustable seats — is now so universal we don't even notice it.

Engineers Make the Same Mistake

Design problem	Average-based approach	What you actually need
HVAC cooling capacity	Size for ~72 °F annual average	Size for the 1% hottest hours
Structural wind load	Design for ~7 mph mean wind	Design for the peak gust pressure
Water management	Base on average rainfall	Base on peak storm intensity
Bridge traffic load	Use average daily truck count	Use the heaviest truck combination

⚠ In every case, a system sized for the average fails when it is stressed most — exactly when it must not fail.

Your Turn — Add a 5th Row

2 minutes · discuss with your neighbor

Think of a case — from civil or architectural engineering — where a system is commonly sized for the average but should be sized for an extreme.

Be ready to share:

- What is the **system**?
- What is the **average-based mistake**, and what should the **design value** actually be?

✓ Your Turn — Sample Answers

Any of these — or your own — is correct

Design problem	Average-based mistake	What you actually need
Stormwater pipe sizing	Size for mean annual rainfall rate	Size for the 25-year design storm intensity
Electrical service panel	Use average household consumption	Use the simultaneous peak demand hour
Foundation bearing	Use average dead + live load	Use peak load combination per ASCE 7
Glazing U-value	Use mean winter temperature	Use ASHRAE 99.6% design cold night

The pattern is the same every time: failure happens at extremes, not averages.

Part 2

Descriptive Statistics

Three questions every engineer asks about data

Three Questions to Ask About Any Dataset

When you look at data for the first time, you need to answer three questions:

1. Center

What is the typical value?

Where does the data cluster?

2. Spread

How much do values vary?

How far do they scatter from the center?

3. Shape

Is the data symmetric?

Are there extremes that matter?

💡 These three questions are the foundation of statistics — and the first step of every AI analysis pipeline.

Predict Before You See the Data

75 seconds · write your guess

We are about to open a dataset of every hour of Tucson weather in 2023 — 8,760 rows.

Before we do: write down your best guess for each of these.

- What is the **annual average temperature** in Tucson?
- What was the **hottest temperature** recorded in 2023?
- How many hours per year do you think **exceeded 100 °F**?

The Dataset — 8,760 Hours of Tucson Weather

Source: NOAA Automated Surface Observing System (ASOS), Tucson International Airport, 2023

Column	What it measures
DateTime	Timestamp — one row per hour
Temp_F	Dry-bulb air temperature (°F)
Wind_Speed_mph	Sustained wind speed (mph)
Relative_Humidity_pct	Relative humidity (%)

8,760 rows = 365 days × 24 hours — every hour of an entire year, off a real instrument.



An ASOS array beside a runway — thermometer, anemometer, ceilometer, precipitation gauge. Every row in your file came off equipment like this.
Photo: Famartin, CC BY-SA 4.0, via Wikimedia Commons.

Center — What the Mean Tells You (and What It Hides)

The **mean** (arithmetic average) is the most common measure of center. It answers: *if all values were equal, what would that value be?*

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

where x_i = each data value, $\bar{x}$ = sample mean, n = sample size

The mean has a critical weakness: **it treats every observation equally** — a heat wave hour and a mild spring morning count the same.

- Tucson 2023 annual mean temperature: **$\approx 72^\circ\text{F}$**
- Tucson also recorded **112°F**
- A system sized for 72°F is undersized by 40°F on the hottest day

The engineering question: does the mean matter here — or the extreme?

Center — The Median

The **median** is the middle value when observations are sorted — half fall below it, half above.

Why it matters: the median is not affected by how extreme the extremes are.

$$\tilde{x} = \begin{cases} x_{(\frac{n+1}{2})} & n \text{ is odd} \\ \frac{x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}}{2} & n \text{ is even} \end{cases}$$

where $x_{(k)}$ = the k -th value in sorted order, n = sample size

	Mean	Median
Sensitive to extremes	Yes	No
Best when	Distribution is symmetric	Data is skewed or has outliers

Example — construction cost overruns:

A few catastrophic projects can pull the mean far above what a typical project costs. The median tells you what a *typical* project actually cost.

Calculating the Median — Step by Step

Odd n — 5 values (°F)

Sort: **28, 58, 73, 86, 112**

$$\text{Position: } \frac{n + 1}{2} = \frac{5 + 1}{2} = 3$$

Median = **73 °F** (3rd value)

💡 The single middle value — no averaging needed.

Even n — 4 values (°F)

Sort: **58, 65, 86, 112**

Middle two: positions 2 and 3

$$\text{Median} = \frac{65 + 86}{2} = \mathbf{75.5\text{ }^{\circ}\text{F}}$$

💡 Average the two middle values when n is even.

Key property: change 112 °F to 500 °F — the mean jumps by 78 °F, the median stays at 73 °F. Position determines the median; magnitude does not.

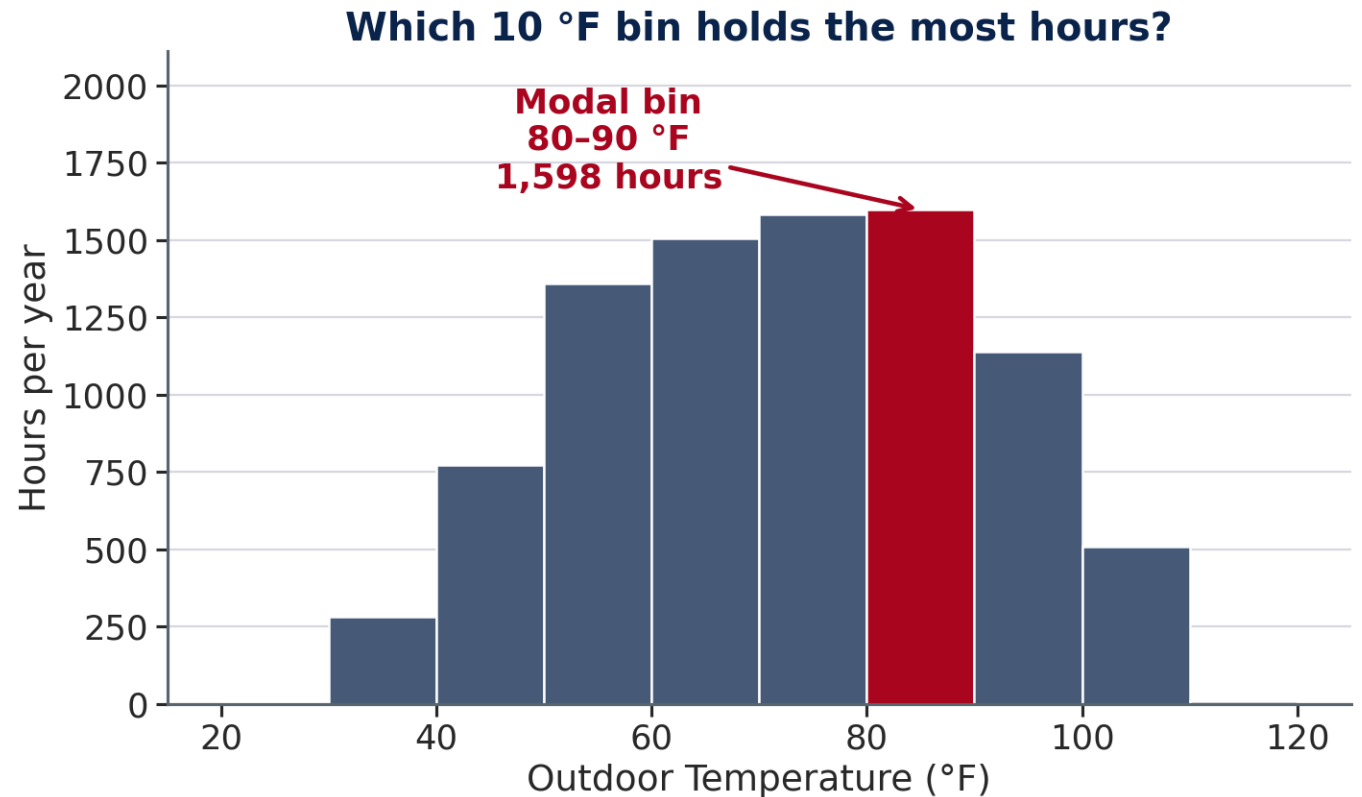
Center — Mode

The **mode** is the most frequently occurring value in a dataset.

For discrete data it is directly useful — most frequent wind direction, most common inspection rating.

For continuous data raw values rarely repeat, so a single "most common value" is meaningless. We **bin** first, then ask which bin holds the most hours.

💡 Exactly what A13 asks for in the exercise this week.



Three Measures of Center — Which One, When?

Measure	What it answers	Reach for it when
Mean	If all values were equal, what would they be?	You need a total or a budget
Median	What is the middle value?	The data is skewed or has outliers
Mode	Which value — or bin — appears most often?	The data is categorical, or you want the busiest range

For symmetric distributions, all three agree: mean $\approx$ median $\approx$ mode.

When they diverge, the distribution is skewed — and that divergence is your first warning that the average may be misleading you.

Spread — Standard Deviation

Standard deviation (s) — the typical distance any observation sits from the mean.

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

where x_i = each data value, $\bar{x}$ = sample mean, n = sample size

Tucson 2023 outdoor temperature:

- $s = 18\text{ °F}$, mean = 72 °F
- A typical hour sits 18 °F from the annual mean
- Typical range: roughly $54\text{--}90\text{ °F}$

Excel formula:

`=STDEV(range)` — divides by $n - 1$, not n
(Bessel's correction — the next two slides explain why)

Use `STDEV` (sample) for almost every real engineering dataset — you have a sample, not the full population.

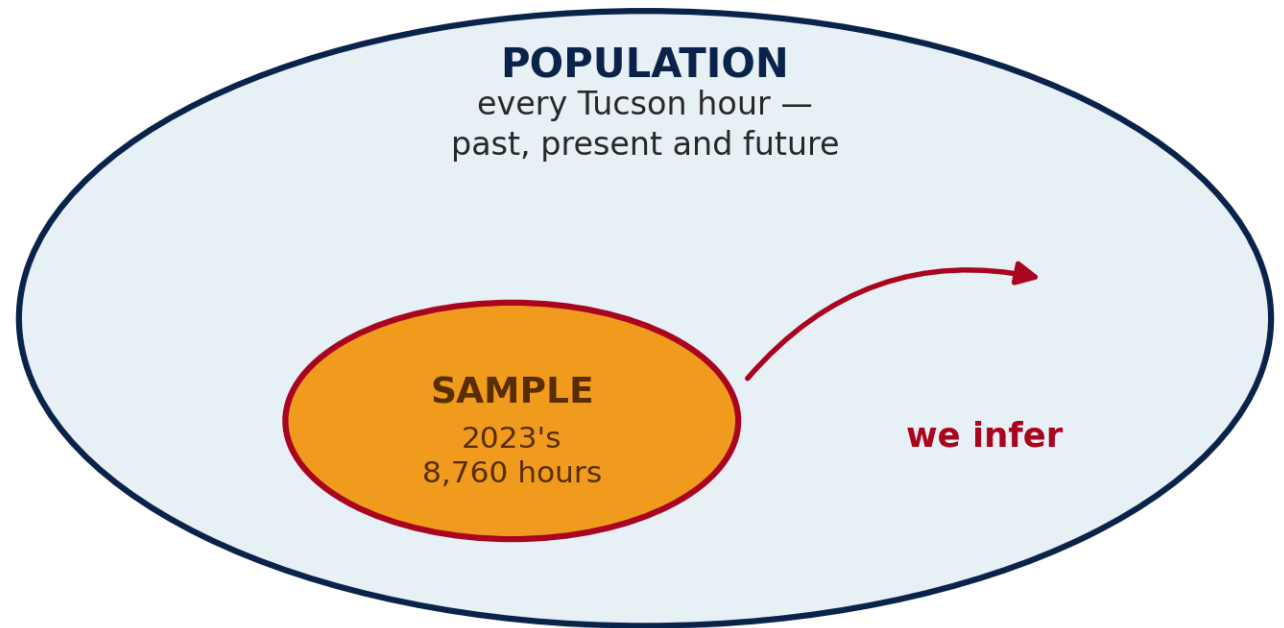
Sample vs. Population — What's the Difference?

That formula divided by $n - 1$, not n — a deliberate choice, and this is the distinction behind it.

Population — every value that could ever exist for what you are studying.

Sample — the data actually in your hands.

You almost never have the population. Every statistic you compute from a sample — mean, SD, quartiles — is an **estimate** of the population value, not the value itself.



*Every statistic you compute is an estimate,
not the value itself.*

Sample vs. Population Standard Deviation

	Formula	When to use
Population σ	Divide by n	You have every value (rare)
Sample s	Divide by $n-1$	You have a subset (almost always)

Why $n-1$? A sample tends to underestimate the true population variability. Dividing by $n-1$ instead of n corrects this — it is called **Bessel's correction**.

Always use STDEV unless you are certain you have the entire population.

💡 Estimating the mean "uses up" one piece of information, leaving only $n-1$ independent pieces to estimate spread — a **degree of freedom**. The term returns in regression (Week 10) and hypothesis testing (Week 11).

Spread — Coefficient of Variation (CV)

Coefficient of variation (CV) = $s \div \text{mean}$ — expresses spread as a fraction of the mean, removing units so you can compare across different variables.

$$CV = \frac{s}{\bar{x}} \times 100\%$$

	s	Mean	CV
Outdoor temp	18 °F	72 °F	25%
Zone temp (<i>Week 5</i>)	2.9 °F	73.5 °F	4%

Same building, same city — very different spread relative to the mean.

Reading CV as a design signal:

- **CV < 10%:** data clusters tightly around the mean — mean is a reliable design value
- **CV 10–20%:** moderate spread — mean needs context; report s alongside it
- **CV > 20%:** high spread — the mean is likely misleading as a standalone design value; use percentiles

⚠ Outdoor temperature: CV = 25%. The mean (72 °F) describes only the center. The system must handle 108 °F peaks — 36 °F above that mean.

Small vs. Large CV — What It Looks Like

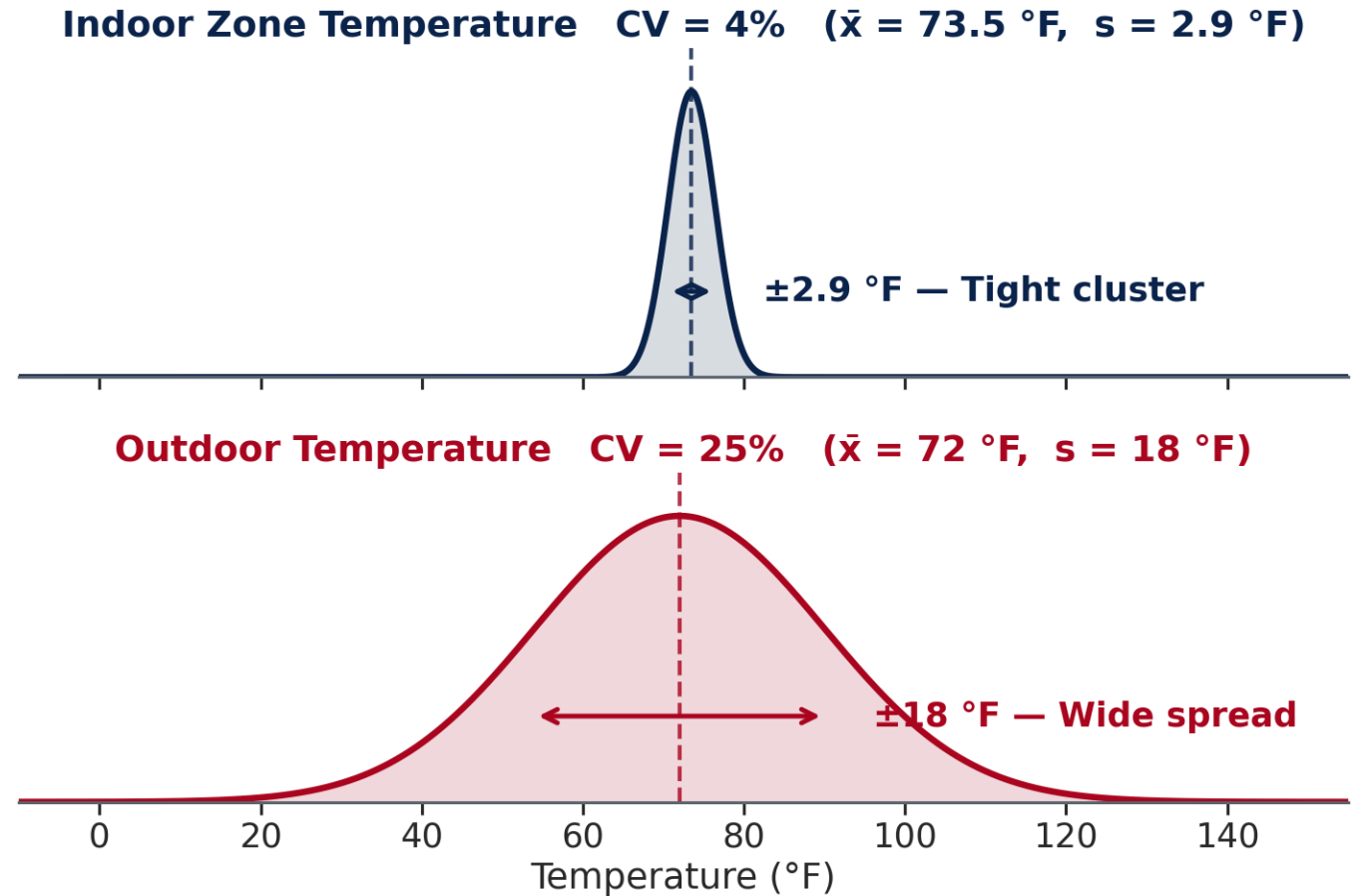
CV = 4% → mean is reliable

Indoor zone temperatures cluster tightly around the setpoint. The mean describes nearly every hour well.

CV = 25% → use percentiles

Outdoor temperatures spread widely across the year. The mean sits at the center but misses the extremes that drive equipment sizing.

Same city, same building — very different spread relative to the mean.



When Does the Mean Work Fine?

The flaw of averages is real — but the mean is not always wrong. Knowing when to trust it matters as much as knowing when not to.

Mean is appropriate when:

- The distribution is roughly symmetric (mean $\approx$ median)
- No single extreme value drives the outcome
- You need a total or balance — energy budgets, material quantities, cost projections
- The consequence of being wrong is low and reversible

Design percentile is required when:

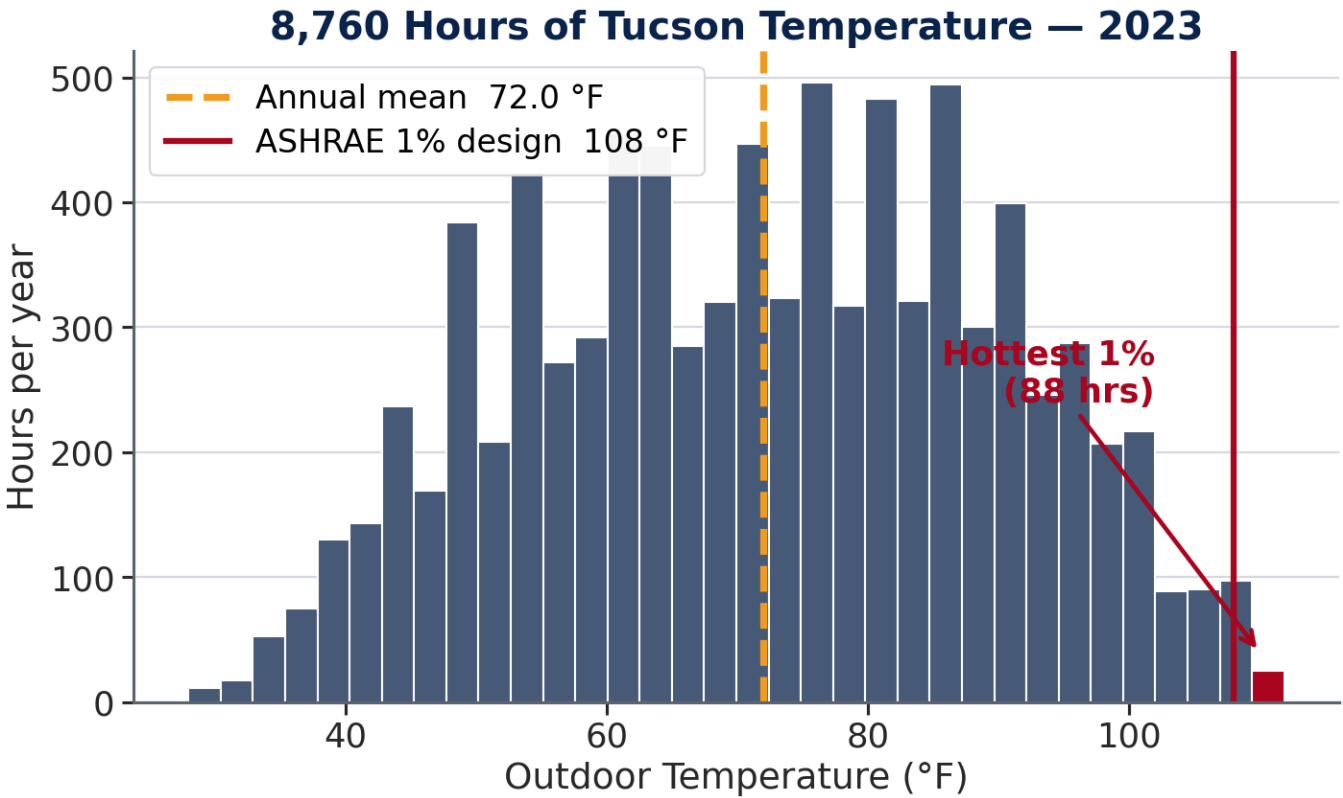
- Failure at an extreme has serious consequences
- The distribution is right-skewed (floods, gusts, peak loads)
- A system must be sized to handle the worst case it will face
- *Energy budget* → mean; *Cooling capacity* → 99th percentile

Tucson 2023 — Hours Above Threshold

How often does Tucson actually exceed design temperatures?

Threshold	Hours per year
Above 95 °F	≈ 900 hrs
Above 100 °F	≈ 429 hrs

429 hours is nearly **18 full days** of continuous operation above 100 °F — not a rare spike. A cooling system sized for the annual average (72 °F) would be undersized for months of every year.



✓ How Did You Do? — Tucson 2023

From your guesses a few slides ago

Your prediction	What the data shows
Annual average temperature	$\approx 72^{\circ}\text{F}$
Hottest recorded temperature	112 °F — July 13, 2023
Hours above 100 °F	≈ 429 hours (~18 continuous days worth)

The gap that matters: the mean is 72 °F — comfortable, unremarkable.

The extreme is 112 °F — **40 °F above the mean**, and the number that sizes your HVAC system.

Most people underestimate how many hours exceed 100 °F. 429 hours is nearly 6 weeks of working hours in a year — not a rare spike.

Part 3

Design Percentiles

Designing for the extreme, not the average

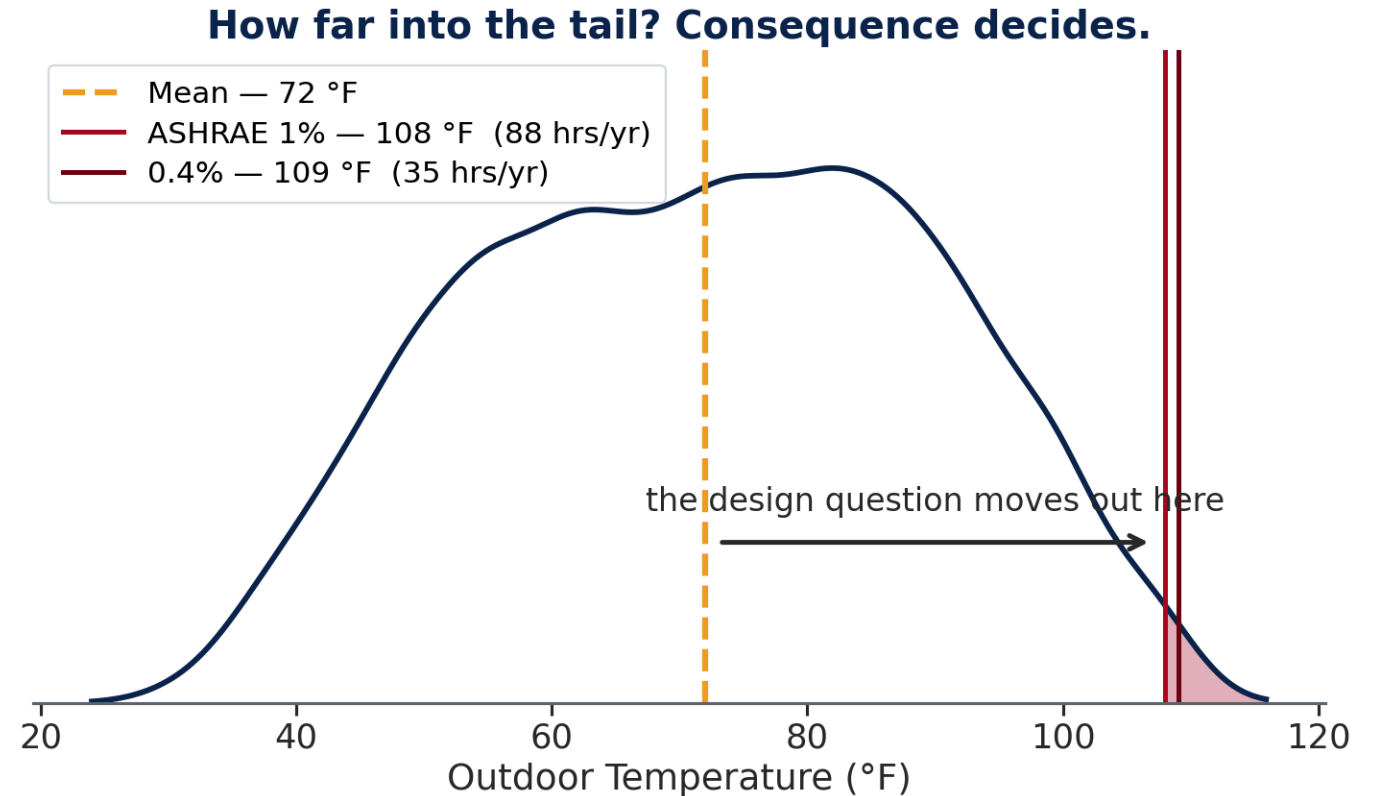
When Does a System Get Stressed?

Not during the average hour — during the **worst** hour.

How far into the tail? The consequence of failure decides:

- **HVAC comfort failure** → discomfort → ASHRAE 1%, ≈ 88 hrs/yr acceptable
- **Structural failure** → loss of life → ASCE 7 return period

Same tool. The **depth** is an engineering judgement, not a lookup.



What Is a Percentile?

Imagine sorting all 8,760 annual temperatures from hottest to coldest.

- **Rank 1** — the single hottest hour of the year ($\approx 112^\circ\text{F}$)
- **Rank 88** — the 88th hottest hour of the year
- **Rank 4,380** — the middle value (the median, 50th percentile)
- **Rank 8,760** — the coldest hour of the year ($\approx 28^\circ\text{F}$)

A **percentile** is simply a rank expressed as a fraction of the total.

ASHRAE 1% cooling design temperature = the temperature at rank 88:

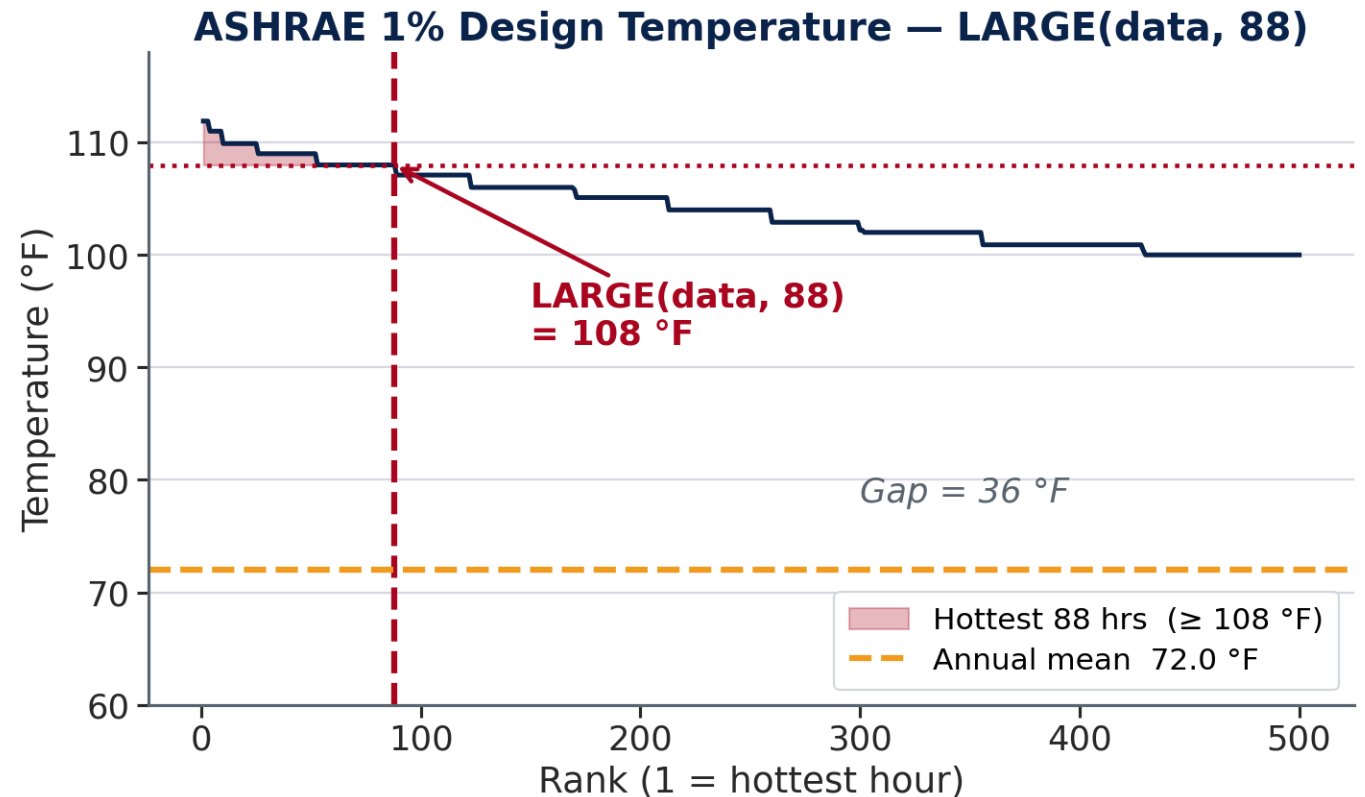
- $1\% \times 8,760 = \mathbf{88 \text{ hours}}$
- Size your cooling system for this temperature $\rightarrow$ it works for **99% of all hours**

The ASHRAE 1% Design Temperature — 108 °F

All 8,760 hours sorted from hottest to coldest. The top 88 hours (1% × 8,760) are highlighted in red.

The temperature at rank 88 is the **ASHRAE 1% design temperature: 108 °F.**

Size your cooling system to handle this temperature → it performs correctly for 99% of all annual hours.



ASHRAE = American Society of Heating, Refrigerating and Air-Conditioning Engineers · 1% design temperature = threshold exceeded in only 88 hrs/yr · the US standard for HVAC equipment sizing

Design Temperature vs. Annual Average

Statistic	Value
Annual average temperature	$\approx 72\text{ }^{\circ}\text{F}$
ASHRAE 1% design temperature	$\approx 108\text{ }^{\circ}\text{F}$
Gap	$\approx 36\text{ }^{\circ}\text{F}$

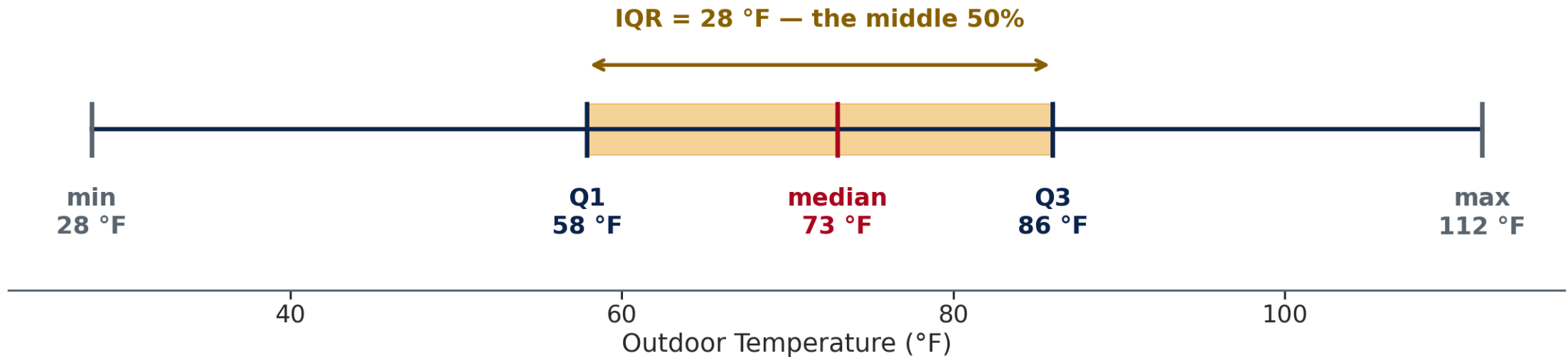
A cooling system sized for the annual average would be **undersized by 36 °F** on the hottest days.

💡 A 36 °F gap between the average and the design temperature is not just a number — it represents a dramatic undersizing of any system designed for the mean. Part 5 (Application 1) quantifies exactly how large that difference becomes in terms of required cooling capacity.

Spread — Quartiles and IQR

Quartiles are **percentiles at 25% and 75%** — the same concept as the ASHRAE design temperature, applied to the middle of the distribution instead of the tail.

Quartiles split the sorted year into four equal parts



Why IQR matters: unlike the range — 28 °F to 112 °F, set entirely by two single hours — the IQR describes the spread of the typical half of the data. It is robust to outliers.

Quick Vote — January or July?

No peeking at the data · raise your hand

We know:

- **January** average ≈ 51 °F — cool, dry winter
- **July** average ≈ 94 °F — hot monsoon season

Which month has **more day-to-day temperature variability**?

- A)** January — cold fronts and warm spells create wide swings
- B)** July — the monsoon creates unpredictable conditions
- C)** About the same — high average = high variability

Part 4

Conditional Statistics

Asking smarter questions of your data

The Structure Hidden Inside the Annual Average

When you compute the annual average, you combine all 8,760 hours — January and July together.

But January and July are completely different conditions for an engineer. Combining them into one number erases that structure.

Conditional statistics answer a more specific question:

What is the average — for a specific subset of the data?

Instead of: "What is the annual average temperature?"

Ask: "What is the average temperature **in July**?"

Or: "How variable is temperature **in winter**?"

Slicing the data by a condition — month, time of day, building type, occupancy state — reveals patterns the overall average hides.

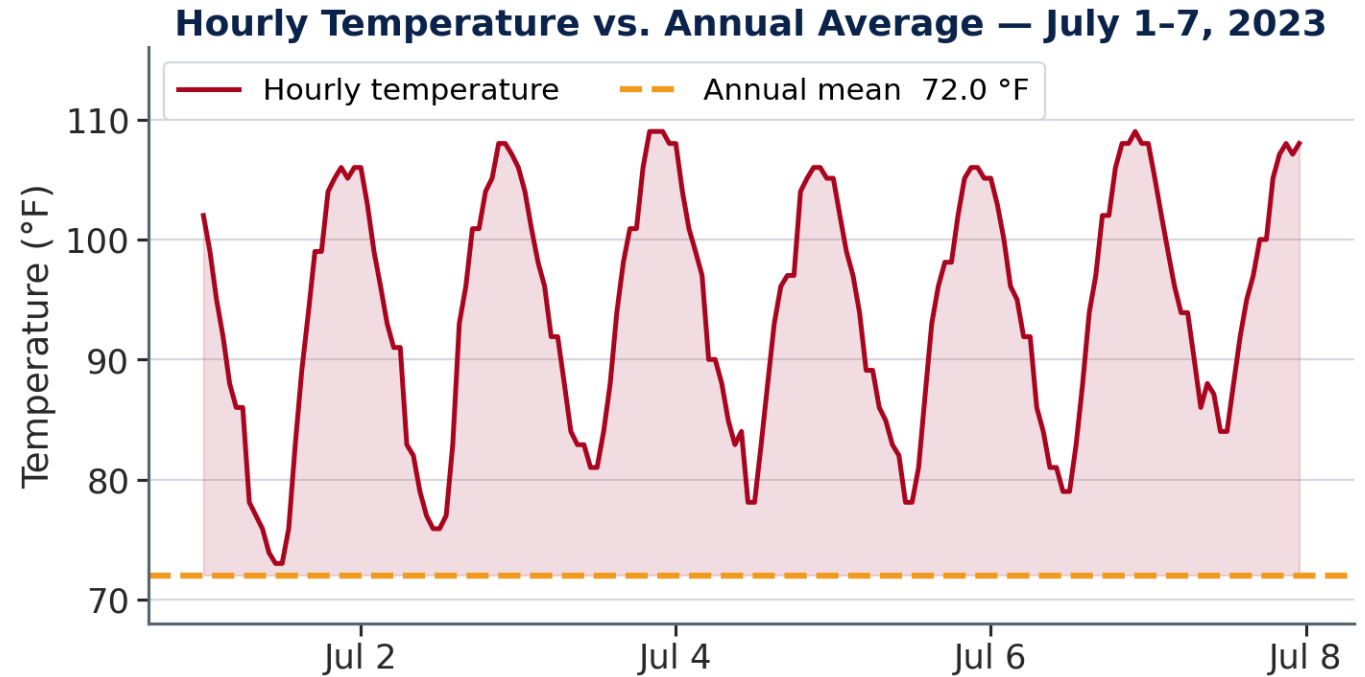
Not All Hours Are Equal

The annual average hides seasonal structure.

- January avg $\approx 51^{\circ}\text{F}$
- July avg $\approx 94^{\circ}\text{F}$
- Gap: 43°F

Every July hour is above the annual mean (gold line). The pink fill is the cooling load the average misses.

Conditional statistics ask: *what is the mean for a specific subset?*

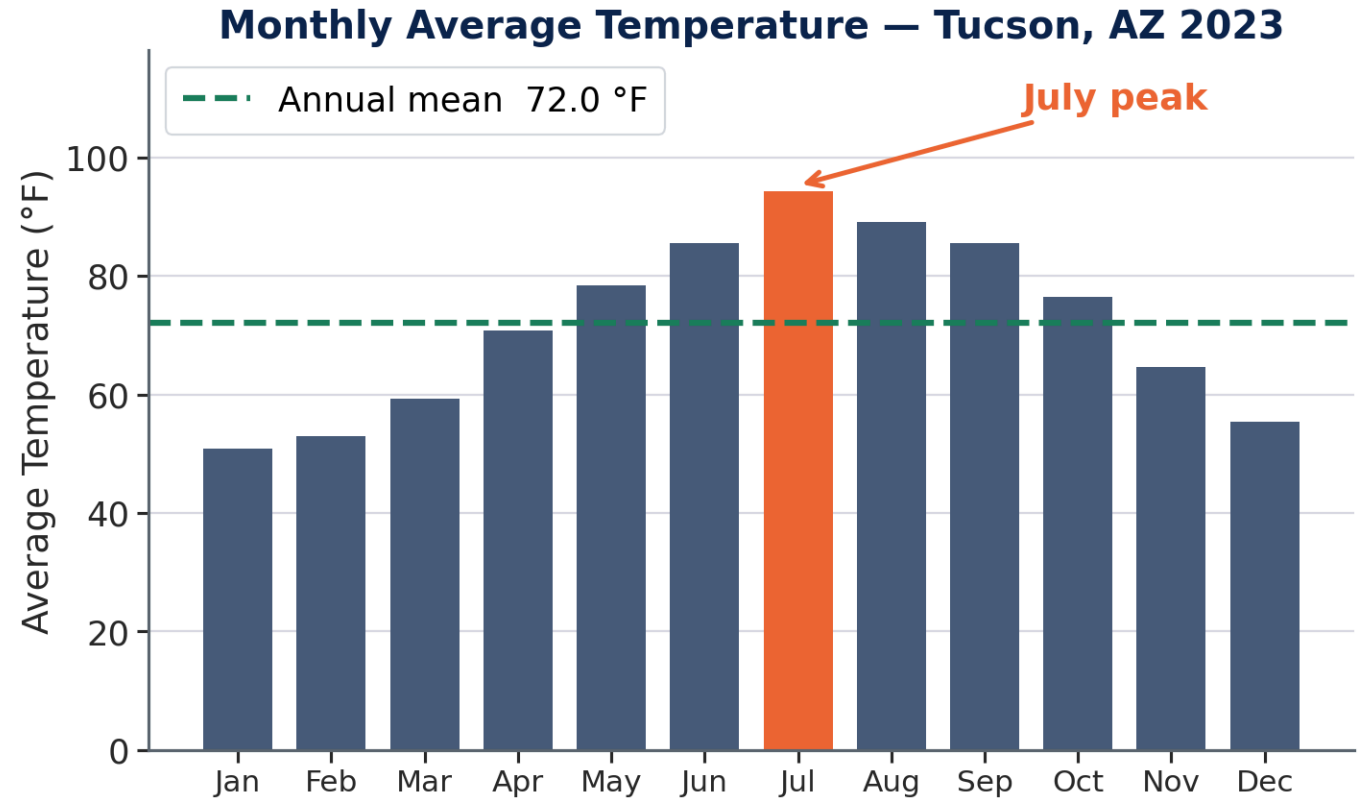


Monthly Temperatures — The Hidden Structure

Conditioning on month reveals what the annual average hides.

Month	Average temp
January	51 °F (43° below July)
July	94 °F (22° above annual mean)

5 months sit above the annual mean of 72 °F; 7 sit below. The distribution of monthly averages is itself right-skewed — the summer heat cluster pulls the mean up.



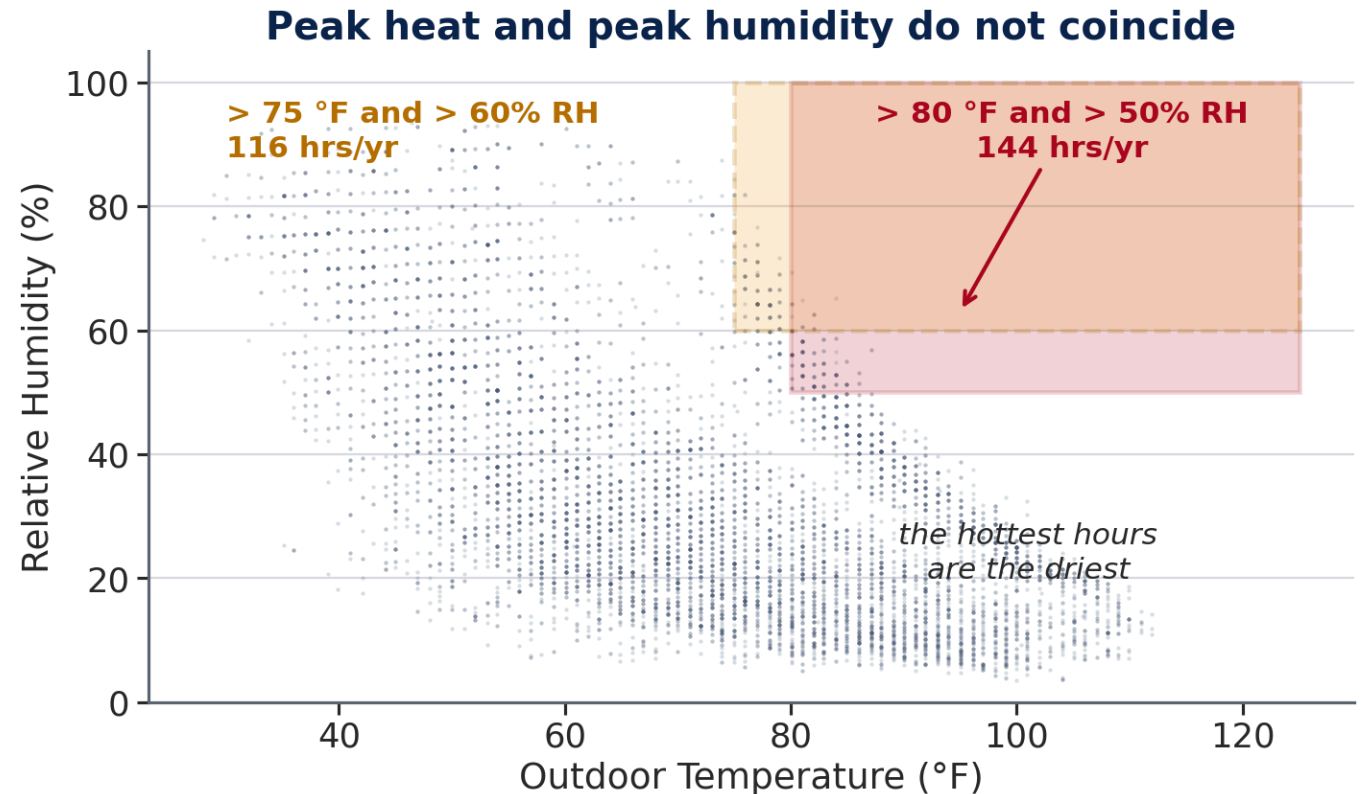
Monsoon Comfort Burden – Two Conditions

Peak heat and peak humidity do not coincide in Tucson.

Condition	Hours/yr
T > 80 °F and RH > 50%	144
T > 75 °F and RH > 60%	116

Tucson's hottest afternoons stay dry.
Monsoon humidity peaks in the evening, after temperatures drop — so the worst comfort hours are warm, humid **evenings**.

💡 Sizing for peak temperature alone misses the latent burden.

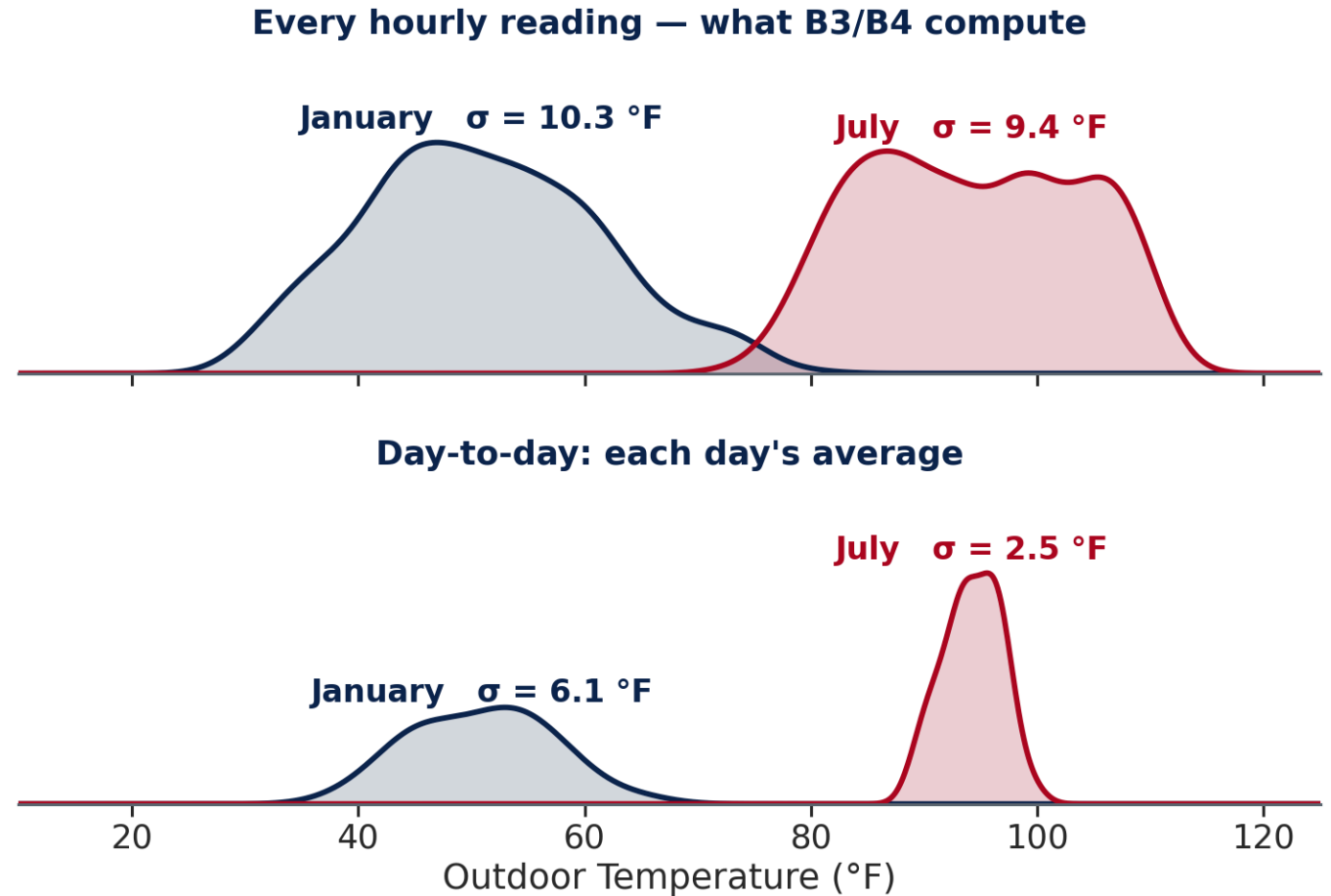


Conditional Variability — January vs. July

Month	Mean	σ , hourly	σ , day-to-day
January	51 °F	10.3 °F	6.1 °F
July	94 °F	9.4 °F	2.5 °F

By hourly readings the two months look almost alike. By day-to-day spread, January is 2.4× as variable.

Why: July's ≈ 26 °F day–night swing repeats nearly identically every day. January's *days* genuinely differ — 29 °F to 80 °F.





Quick Vote — The Answer Was A: January

Here's what the data says

Why? July's daily averages barely move — 89 °F to 99 °F all month, because the monsoon repeats the same cycle every day. January's days genuinely differ, swinging between cold fronts and warm spells.



Key principle: average and variability are independent. January is colder on average *and* far less predictable day to day.

Watch the definition. Take the spread over every *hour* instead and the two months are nearly tied — 10.3 °F vs 9.4 °F — because July's fixed 26 °F day–night swing inflates its hourly spread without making the month unpredictable. **"More variable" is not a question you can answer until you say what you are averaging over.** You will compute the hourly version yourself in B3 and B4 this week.

Part 5

Three Engineering Applications

Application 1 — HVAC Cooling (ArcE / ME)

Design question: What cooling capacity does a Tucson office building require?

Average-based (wrong):

Outdoor temp $\approx 72^\circ\text{F}$ · Indoor setpoint = 75°F

$\Delta T = 3^\circ\text{F} \rightarrow$ very small cooling load

Design-percentile approach (correct):

ASHRAE 1% design temp = 108°F

$\Delta T = 33^\circ\text{F} \rightarrow$ **11× larger load**

Size the chiller for this demand

 A chiller sized for the annual average fails on every peak summer afternoon — roughly 429 hours above 100°F per year. That is 18 full days of system failure.

Application 2 — Wind Pressure (CE)

ASCE 7-22 (*ASCE Minimum Design Loads Standard — the US code for wind, seismic, and snow loads*)

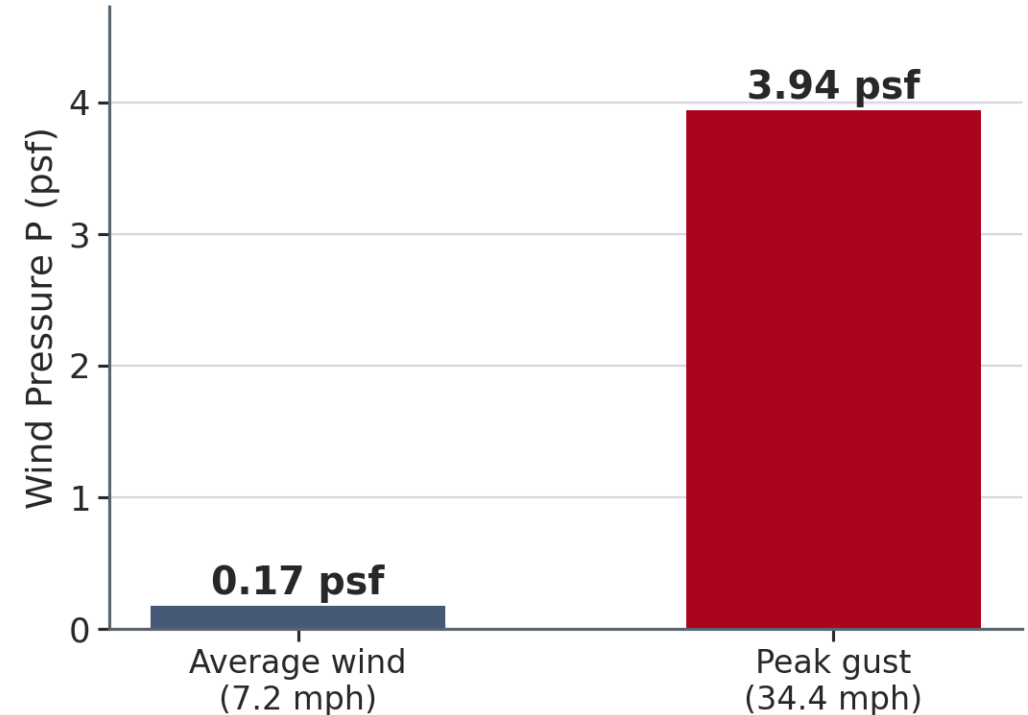
$P = 0.00256 \times K \times V^2$, where $K = 1.3$ combines exposure and gust

Peak gust is only **4.7× faster** — but V^2 means **23× more pressure**.

$$\frac{P_{peak}}{P_{avg}} = \left(\frac{34}{7.2} \right)^2 \approx 23$$

A billboard sized for average wind sees **23× its design load** at peak gust.

Peak is 23× Average
 $P = 0.00256 \times 1.3 \times V^2$ (ASCE 7-22)



💡 Any V^2 -dependent quantity (wind, seismic, head loss) amplifies the average-vs-peak gap.

Application 3 — Latent Load / Humidity (ArcE / ME)

ASHRAE 62.1 requires HVAC systems to remove moisture from outdoor ventilation air — on top of temperature control.

As seen in the Part 4 Monsoon slide: Tucson records **144 hrs/yr** above 80 °F and 50% RH, and **116 hrs/yr** above 75 °F and 60% RH — despite its desert identity.

Mold risk rises sharply above 60% RH. These monsoon hours require **dehumidification capacity** that a temperature-only analysis misses entirely.

Design rule: size the dehumidification coil for the peak ventilation moisture load — not the annual average RH ($\approx 45\%$).

💡 A building that maintains the correct temperature but no moisture control will grow mold during every monsoon season — regardless of how well the sensible cooling system is sized.

Part 6

In-Class Exercise This Week · Week 2 Preview

What You Will Do in the Exercise — Week 1

Dataset: `Week1_Tucson_Weather_2023.csv` — 8,760 rows, 4 columns · load into Excel

Section A — mean, median, max, min, range, σ , Q1/Q3/IQR, exceedance counts, hottest month

Section B — ASHRAE 1% design temperature (B1–B2) · conditional σ by month (B3–B6) · wind pressure (B7–B11, CE) · humidity exceedance (B12–B14, ArcE)

Memo — 3 paragraphs: averages problem · design temperature · latent cooling burden

Pacing: Section A + B1–B6 + Chart 1 in class · B7–B13, Chart 2, memo at home

Answer Template Rule

Your answers are read from **specific cells in a specific file**.

⚠ Do not rename the file, change the sheet tab, move rows, or put answers in the wrong column. Answers outside the designated cell receive zero — no manual re-grading after the deadline.

Formulas go in Col C as text — put an apostrophe before the `=`, e.g. `'=AVERAGE(B2:B8761)`. The formula itself, not "computed in Excel": Col C is matched against the method the question asks for, and a missing or wrong formula caps that question at two-thirds of its points. Screenshots are not accepted.

No Deadline Extension Rule:

- After the deadline, you will not get a chance to submit or re-submit, no exceptions.

Three Deliverables This Week

	What it covers	When
Weekly Quiz	15 questions on this lecture and the Tutorial	Closes 8:00 AM Tuesday, Sep 1
In-Class Exercise	Sections A and B, the charts, and the memo	Work Wed + Fri · due Wed 9:00 AM next week
Homework	A separate take-home problem set	On your own · due Wed 9:00 AM next week

The exercise spans two sessions. You start it Wednesday after the tutorial, and finish it Friday. **You are not expected to finish in class** — use the session time for the parts where having me in the room actually helps, and complete the rest on your own.

💡 Three submissions, three D2L Assignments folders. The quiz closes 8:00 AM Tuesday, Sep 1; the exercise and the homework are both due Wednesday 9:00 AM — no late work is accepted.

Week 2 Preview — Distributions

You will compute a lot of statistics this week — but statistics alone don't show **shape**.

Think about this before next class:

If you plotted all 8,760 temperature values as a histogram — bins on the X-axis, count of hours on the Y-axis — what shape would you expect?

- Symmetric? Skewed? Bimodal?
- Where would the peak(s) be?
- How does the shape connect to what you already know about Tucson's seasons?

We will build this histogram in Week 2 — and learn how to match the shape to a named probability distribution.

Week 1 Summary

The flaw of averages · designing for the mean fails when extremes drive the outcome

Descriptive statistics · mean · median · mode · range · IQR · σ · skewness · percentiles

Design percentile · ASHRAE 1% $\approx$ 108 °F for Tucson

Conditional statistics · monthly averages · conditional spread · seasonal structure hidden by annual average

Three applications · HVAC sizing · Wind pressure (V^2) · Latent monsoon load

AI connection · these tools are the foundation of every prediction model you will use or build